

The Naylor Model

Variable Glossary

Statcast-style reference · v6

Every variable in this glossary tells you:

- what it actually measures (plain English)
 - the units and a typical value
 - what's elite and what's poor
 - a real player example
 - why it affects steal success

No statistical jargon.

Replace 'coef_z' and 'OR/SD' with
'SB % Boost per Tier' and 'Odds Multiplier'.

Speed & Acceleration

Sprint Speed

What it is

Top running speed measured during the player's fastest sprints.

Units	League avg	Elite
feet per second (ft/s)	27.0	29.0 +

Real example

Bobby Witt Jr.: 30.2 (elite) · Josh Naylor: 24.4 (3rd %ile)

Tier chart

Elite	29.0 +
Above avg	27.5 – 29.0
Average	26.5 – 27.5
Below avg	25.5 – 26.5
Poor	≤ 25.5

Why it matters

Faster runners are obviously harder to throw out, but the model finds it matters LESS than raw acceleration on a 90-ft steal. Once you're above ~28 ft/s the extra speed is wasted because you're past second base before the throw arrives anyway.

Source

Baseball Savant Sprint Speed leaderboard

Speed Capped

What it is

Sprint speed, but anything over 28 ft/s is clipped to 28.

Units	League avg	Elite
ft/s	27.0	28.0 (capped)

Real example

Witt 30.2 → capped at 28.0 · Naylor 24.4 → unchanged

Tier chart

 ≥ 28 ft/s	treated as 28.0 (no extra reward)
 < 28 ft/s	kept as-is — every fraction matters

Why it matters

The model adds this version because the marginal benefit of more speed vanishes above 28 ft/s. Capping prevents the model from over-rewarding pure speedsters and lets it find smaller advantages like jump and lead.

Source

computed from Sprint Speed

Jump Time (accel_0_30)

What it is

How fast you cover the first 30 feet from contact.

Units	League avg	Elite
seconds	1.78	≤ 1.65

Real example

Corbin Carroll: 1.67 · Josh Naylor: 1.87 (slow) · Nasim Nuñez: 1.64 (elite)

Tier chart

 Elite	≤ 1.68
 Above avg	1.69 – 1.74
 Average	1.75 – 1.80
 Below avg	1.81 – 1.86
 Poor	≥ 1.87

Why it matters

The single most important running metric for steals — the throw is in the air before you hit top speed, so what happens in the first 30 ft decides everything. In the v5/v6 simple GLM this feature has the strongest weight of any speed variable.

Source

Statcast Running Splits · seconds_since_hit_030

Acceleration Phase (accel_5_30)

What it is

Time from 5 ft to 30 ft — pure first-burst, excluding bat-flip.

Units	League avg	Elite
seconds	1.32	≤ 1.22

Real example

Witt: 1.23 (elite) · Naylor: 1.38

Tier chart

Elite	≤ 1.22
Above avg	1.23 – 1.28
Average	1.29 – 1.34
Below avg	1.35 – 1.40
Poor	≥ 1.41

Why it matters

Cleaner version of Jump Time — removes how quickly the runner cleared the batter's box. Less noisy proxy of biomechanical first-burst.

Source

Statcast Running Splits · 030 – 005

Top-Speed Phase (maintain_30_90)

What it is

Time from 30 ft to 90 ft — back half of the sprint.

Units	League avg	Elite
seconds	2.27	≤ 2.15

Real example

Witt: 2.10 (elite) · Naylor: 2.44

Tier chart

Elite	≤ 2.15
Above avg	2.16 - 2.22
Average	2.23 - 2.30
Below avg	2.31 - 2.38
Poor	≥ 2.39

Why it matters

Matters less for SBs than Jump Time — a steal of 2nd only requires covering ~85 ft. But it captures conditioning and shows up in pre-2023 data as a stronger predictor than today.

Source

Statcast Running Splits · 090 - 030

Total 90 (total_90)

What it is

Full home-to-first time from contact.

Units	League avg	Elite
seconds	4.05	≤ 3.95

Real example

Carroll: 3.94 (elite) · Naylor: 4.31 (15th percentile)

Tier chart

Elite	≤ 3.95
Above avg	3.96 - 4.02
Average	4.03 - 4.10
Below avg	4.11 - 4.18
Poor	≥ 4.19

Why it matters

A summary number — equals Jump Time + Top-Speed Phase. Useful for ranking 'overall fast' runners. Naylor's 4.31 puts him in the bottom 25%, yet his accel_gap (below) shows he's still quick OFF THE LINE — the steal-relevant part.

Source

Statcast Running Splits · seconds_since_hit_090

Accel Gap

What it is

How much faster the runner is off the line than their top speed implies.

Units	League avg	Elite
percentile points	0	+ 25 or more

Real example

Naylor: +22 to +30 across seasons · Witt: -20 (slower off the line for his speed)

Tier chart

 Naylor archetype	+15 or more (fast jump despite low speed)
 Above expected	+5 to +15
 As expected	-5 to +5
 Below expected	-15 to -5
 Pure speedster	≤ -15 (fast top speed but slow start)

Why it matters

This is the heart of the Naylor archetype. POSITIVE values mean the runner is faster off the line than their top speed would predict — exactly what a slow-but-effective stealer looks like. Pure speedsters often have NEGATIVE accel_gap because they coast on top-end speed.

Source

computed: `pctile(Jump Time, inverted) - pctile(Sprint Speed)`

Bolts

What it is

Number of plays in a season where the runner topped 30 ft/s.

Units count	League avg 13	Elite 100 +
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Real example

Witt: 250+ · Carroll: 200+ · Naylor: 0 (never hit 30 ft/s)

Tier chart

 Elite	100 +
 Above avg	30 - 100
 Average	5 - 30
 Below avg	1 - 5
 Never	0

Why it matters

A simple count of elite-speed moments. Big leaderboard presence; the model treats it as a speed credential. Zero bolts doesn't disqualify you (Naylor proves that), but it puts you in the slow-steal archetype the model is designed to identify.

Source

Baseball Savant

Lead Distance (real Baseball Savant)

Primary Lead

What it is

Average distance off first base when the pitcher starts his motion.

Units	League avg	Elite
feet (ft)	11.5	13.0 +

Real example

Trea Turner: 13.1 (elite) · Mookie Betts: 12.4 · Naylor: 9.8 (small lead)

Tier chart

 Aggressive	13.0 +
 Above avg	12.0 - 13.0
 Average	11.0 - 12.0
 Conservative	10.0 - 11.0
 Very small	≤ 10.0

Why it matters

Bigger lead = less distance to cover. But coaches push leads to where the pitcher gets nervous, so big leads can also increase pickoff risk. Notably Naylor has one of the SMALLEST primary leads in MLB (9.8 ft) yet steals 90%+ — his edge is elsewhere.

Source

Savant Basestealing Run Value · `r_primary_lead` (career snapshot — Savant ignores year filter on this

Secondary Lead

What it is

Average distance off first base at the moment of pitch release.

Units	League avg	Elite
feet (ft)	14.5	16.0 +

Real example

Mookie Betts: 18.1 (elite) · Naylor: 13.7

Tier chart

 Elite	16.0 +
 Above avg	15.0 – 16.0
 Average	14.0 – 15.0
 Below avg	13.0 – 14.0
 Poor	≤ 13.0

Why it matters

Primary lead PLUS ground covered while the pitcher is delivering. Big secondary leads compress the steal distance AND give the runner momentum.

Source

Savant · `r_secondary_lead` (career snapshot)

Lead Gain (the 'jerk' metric)

What it is

Distance covered between pitcher's first move and pitch release.

Units	League avg	Elite
feet (ft)	3.5	4.5 +

Real example

Mookie Betts: 5.6 (elite — reads pitchers) · Naylor: 3.9 (above avg)

Tier chart

 Elite	4.5 +
 Above avg	3.8 – 4.5
 Average	3.2 – 3.8
 Below avg	2.6 – 3.2
 Poor	≤ 2.6

Why it matters

The user's 'jerk' metric. Captures TWO things at once: how well the runner READS the pitcher's first move AND how explosively they take their secondary lead. Big lead-gain is a hallmark of smart baserunners (Betts, Chisholm Jr., Frazier, Arraez).

Source

Savant · $r_secondary_lead - r_primary_lead$

v5 / v6 NEW Metrics

Pre-Release Velocity (pre_rel_vel)

What it is

How much ground the runner covers per second between pitcher first move and pitch release.

Units	League avg	Elite
ft / sec	2.7	3.5 +

Real example

Jazz Chisholm Jr. 2025: 4.34 (elite) · Naylor: 2.98 · Soto: 2.39

Tier chart

 Elite	3.5 +
 Above avg	3.0 – 3.5
 Average	2.5 – 3.0
 Below avg	2.0 – 2.5
 Poor	≤ 2.0

Why it matters

Even if a pitcher has a fast delivery, a runner who covers a lot of ground in that small window is still dangerous. We use a league-constant delivery (1.30 s) as the divisor because per-pitch pitcher TTP isn't publicly available — so variation is driven by Lead Gain. The leaderboard separates 'smart' baserunners from pure speedsters.

Source

Lead Gain $\div$ 1.30 s

Post-Release Distance (post_rel_dist)

What it is

How much ground the runner covers during the catcher's pop time.

Units	League avg	Elite
feet (ft)	51.5	58.0 +

Real example

Victor Scott II 2025: 59.5 (elite) · Naylor: 41.5 (last %ile)

Tier chart

 Elite	58.0 +
 Above avg	54.0 – 58.0
 Average	50.0 – 54.0
 Below avg	46.0 – 50.0
 Poor	≤ 46.0

Why it matters

After the pitcher releases the ball, the runner has ~1.95 seconds (the catcher's pop time) to cover as much ground as possible before the throw arrives. Slower runners still accelerating during this window cover LESS than sprint × pop implies — we subtract an acceleration tax for them. This is the speedster-archetype metric: Witt, Carroll, Scott II dominate.

Source

Sprint Speed × catcher pop_2b_sba – acceleration correction

3-2 Count Attempt Share (pct_in_HL)

What it is

Share of a runner's stolen-base attempts that came on a 3-2 (full) count.

Units fraction (0.0 - 1.0)	League avg ≈ 0.55	Elite varies
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Real example

Naylor 2023: 100% · Naylor 2025: 0% · Most runners: 50-80%

Tier chart

 Heavy 3-2 stealer	0.75 +
 Mostly 3-2	0.50 - 0.75
 Balanced	0.25 - 0.50
 Rarely 3-2	0.05 - 0.25
 Never 3-2	0.00

Why it matters

WHAT THIS REALLY MEASURES — there's a Statcast quirk: when we parse SB attempts from play descriptions, the count attached is the AT-BAT's FINAL count, not the count when the runner actually broke for second. So an SB that happens mid-AT-BAT where the AB ended on 3-2 looks like 'an SB on a 3-2 count' even if it really happened on 0-0. This means pct_in_HL is really 'fraction of this runner's attempts where the AB happened to end on 3-2'. It carries a small signal but mostly reflects how often the at-bat went to full count. v6 plan: fix by pulling MLB Stats API play-by-play.

Source

regex on Statcast `des` field

Real Stolen-Base Performance

What it is

Hard counts: stolen bases and caught stealings in the season.

Units count	League avg varies	Elite 30 SB +
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Real example

Naylor 2025: 30 SB / 2 CS = 93.8% success

Tier chart

 Elite volume	30 + SB
 Above avg	20 - 30 SB
 Average	10 - 20 SB
 Low	< 10 SB (excluded from model)

Why it matters

Ground truth. We require $SB + CS \geq 10$ in a season for the runner-season to qualify (smaller samples are too noisy).

Source

MLB Stats API

Real SB % (shrunk)

What it is

Stolen-base success rate, adjusted to prevent tiny samples from looking 100%.

Units	League avg	Elite
fraction (0-1)	0.78	0.85 +

Real example

Naylor 2025: 0.917 (30 SB, 2 CS)

Tier chart

 Elite	0.88 +
 Above avg	0.80 – 0.88
 Average	0.72 – 0.80
 Below avg	0.65 – 0.72
 Poor	≤ 0.65

Why it matters

Shrinkage: a 1-for-1 runner doesn't look like 100%. Formula: $(SB + 5 \cdot \text{league\%}) \div (SB + CS + 5)$. League-mean prior 'pulls' small samples toward 78%.

Source

computed

Expected SB %

What it is

The SB% you'd expect from this runner based ONLY on their speed.

Units fraction (0-1)	League avg $\approx$ Real SB %	Elite n/a
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Real example

Naylor 2025 expected: 0.79 (3rd %ile speed) · actual: 0.92

Tier chart

 **Used as a baseline** compare against Real SB% to get residual

Why it matters

Baseline. Subtracting from Real SB% gives the 'speed-adjusted skill' signal.

Source

2nd-order polynomial fit on Sprint Speed across qualified runners

SB Residual (the KEY signal)

What it is

How much better (or worse) the runner is than their speed predicts.

Units	League avg	Elite
fraction (typically -0.20 to $+0.20$)	0	$+0.10$ or more

Real example

Naylor 2025: $+0.122$ (huge overperformer) · Soto 2025: $+0.128$

Tier chart

	Massive overperformer	$+0.10$ +
	Above expectation	$+0.04$ to $+0.10$
	As expected	-0.04 to $+0.04$
	Below expectation	-0.10 to -0.04
	Far below	≤ -0.10

Why it matters

POSITIVE values mean the runner steals MORE successfully than their raw speed suggests. This is the Naylor / Soto story. We weight it heavily in the SSSI index.

Source

Real SB% - Expected SB%

Battery Context (real per-attempt)

Catcher Pop Time (pop_2b_sba)

What it is

Catcher's time from glove to second base on a steal attempt.

Units	League avg	Elite
seconds	1.95	≤ 1.85

Real example

Patrick Bailey: 1.85 (elite) · league avg: 1.95

Tier chart

Elite arm	≤ 1.85
Above avg	1.86 – 1.92
Average	1.93 – 1.97
Below avg	1.98 – 2.04
Poor	≥ 2.05

Why it matters

REAL per-catcher-per-year data starting 2018. Low pop time = throw arrives at 2B sooner = harder to steal. In Model A this is one of the top 5 features.

Source

Statcast Catcher Poptime leaderboard

Catcher Arm Strength (maxeff_arm_2b_3b_sba)

What it is

Maximum-effort throw velocity to second base.

Units	League avg	Elite
mph	84.0	88.0 +

Real example

Korey Lee 2024: 88.3 mph (elite)

Tier chart

 Elite	88.0 +
 Above avg	85.0 – 88.0
 Average	82.0 – 85.0
 Below avg	79.0 – 82.0
 Poor	≤ 79.0

Why it matters

Component of Pop Time. Some catchers have weak arms but quick transfers; some have rockets but slow exchanges. v5/v6 use both as separate features.

Source

Statcast Catcher Poptime

Catcher Exchange (exchange_2b_3b_sba)

What it is

Time from glove receive to throwing-hand release.

Units	League avg	Elite
seconds	0.64	≤ 0.60

Real example

Patrick Bailey 2024: 0.60

Tier chart

Elite	≤ 0.60
Above avg	0.61 – 0.63
Average	0.64 – 0.66
Below avg	0.67 – 0.69
Poor	≥ 0.70

Why it matters

Fast exchange compensates for a weak arm and vice-versa.

Source

Statcast Catcher Poptime

Pitcher Lead Allowed

What it is

Average lead distance runners take against THIS pitcher (career).

Units	League avg	Elite
feet (ft)	3.5 (lead gain allowed)	How good pitcher is great at suppressing lead

Real example

Best at holding runners → low value; Slow to plate → high value

Tier chart

Suppresses leads	≤ 2.8
Above avg	2.8 – 3.3
Average	3.3 – 3.7
Below avg	3.7 – 4.1
Gives up leads	≥ 4.1

Why it matters

Career-aggregate. When this number is small, the pitcher is good at holding runners. Big = runners feast. Joined to each SB attempt for matchup-level modelling.

Source

Savant Pitcher Running-Game Run Value

Pitcher TTP (Time-to-Plate)

What it is

Seconds from pitcher's first motion to ball-in-catcher's-glove.

Units	League avg	Elite
seconds	1.30	≤ 1.15 (slide-step)

Real example

league constant used in v5/v6 (real per-pitch not published)

Tier chart

 Slide step	≤ 1.15
 Fast	1.16 - 1.25
 Average	1.26 - 1.35
 Slow	1.36 - 1.45
 Very slow	≥ 1.46

Why it matters

Real per-pitch TTP would be the single most useful new data source — but Statcast doesn't publish it. Savant's pitch-tempo CSV is buggy at source (the 'with-runners-on' column equals the 'bases-empty' column for every row). v6 substitutes the league constant 1.30 s.

Source

league constant 1.30 (v6)

Game Situation

Inning

What it is

Which inning the play occurred.

Units	League avg	Elite
integer 1 - 15	—	—

Real example

Late-game (7+) shows different attempt rates than early.

Tier chart

Early	1 - 3
Middle	4 - 6
Late	7 - 9
Extras	10 +

Why it matters

Surprisingly important in Model A — the highest SHAP feature. Late-inning leverage changes manager decisions and runner risk tolerance.

Source

Statcast pitch-level

Outs (outs_when_up)

What it is

Number of outs in the half-inning (0, 1, or 2).

Units	League avg	Elite
0 / 1 / 2	—	—

Real example

Two-out steals are riskier; managers send less often.

Tier chart

 0 outs	highest attempt rate
 1 out	average
 2 outs	lowest — risk of ending inning

Why it matters

Used as a feature; not a huge signal but small reliable effect.

Source

Statcast pitch-level

Balls / Strikes (count)

What it is

Number of balls / strikes BEFORE the current pitch.

Units	League avg	Elite
integers 0-3 / 0-2	—	—

Real example

3-2 (full count) is by far the most common SB-attempt context.

Tier chart

 Full count 3-2	highest by far (1.9% per pitch)
 Two-strike	0.27 - 0.29%
 All other counts	≈ 0% (data quirk — see 3-2 share above)

Why it matters

Feeds into 3-2 Count Attempt Share. See limitation note in that entry — exact count at SB is not in `des` text.

Source

Statcast pitch-level

Composite Scores

SSSI v5 (Slow-Steal Skill Index)

What it is

A weighted composite score that identifies slow-but-effective stealers.

Units	League avg	Elite
z-score units (typically −3 to +3)	0	+2.0 or more

Real example

Naylor 2025: +1.99 · Soto 2025: +1.26 · league mean: 0

Tier chart

Elite slow-steal	+2.0 +
Above avg	+1.0 to +2.0
Average	−0.5 to +1.0
Below avg	−1.5 to −0.5
Poor	≤ −1.5

Why it matters

Weighted average of eight standardized features (see v5 report). Designed to find the Naylor / Soto archetype: elite-performing slow runners. Weights were tuned on 80% of runners with Naylor + Soto HELD OUT to avoid overfitting.

Source

computed in v5/v6 pipeline

How to Read the Model's Weight Table

SB % Boost per Tier (standardised logit coefficient)

What it is

If a runner improves this feature by 1 tier (1 standard deviation), how much does the model's predicted SB success rate go up?

Units	League avg	Elite
percentage points (pp)	0	+5 pp or more = a strongly helpful feature

Real example

Jump Time has -10.6 pp → improving Jump Time by 1 tier raises predicted SB% from 62% to 51% (-10.6 pp). Wait, that's WORSE? No — Jump Time is in seconds, and LOWER is better. So a 1-tier INCREASE (slower) hurts by 10.6 pp. The model shows direction with a sign.

Tier chart

 Strongly helps	+5 pp +
 Helps a little	+1 to +5 pp
 Neutral	-1 to +1 pp
 Hurts a little	-5 to -1 pp
 Hurts a lot	≤ -5 pp

Why it matters

Replaces the old 'coef_z' column. Plain-English version of the same number.

Compute as:

$P_{\text{baseline}} = \text{sigmoid}(\text{intercept})$

$P_{\text{after}} = \text{sigmoid}(\text{intercept} + \text{coef})$

$\text{boost_pp} = (P_{\text{after}} - P_{\text{baseline}}) \times 100$

Source

logistic regression coefficient × baseline probability slope

Odds Multiplier (odds ratio per SD)

What it is

If a runner improves by 1 tier, what happens to their ODDS of success?

Units	League avg		Elite
multiplicative factor	1.0	1.20 +	= feature multiplies odds by 20

Real example

Lead Gain has Odds Multiplier 1.16 → improving by 1 tier multiplies the runner's success-odds by 1.16.

Jump Time has Odds Multiplier 0.63 → 1 tier SLOWER cuts the odds to 63% of baseline.

Tier chart

 Strongly helps	1.20 +
 Helps	1.05 - 1.20
 Neutral	0.95 - 1.05
 Hurts	0.80 - 0.95
 Strongly hurts	≤ 0.80

Why it matters

Replaces the old 'OR/SD' column. Multiply with current odds to get new odds.
Equivalent to $e^{\text{coefficient}}$.

Source

`exp(logistic coefficient)`

Understanding the Results

The Naylor Model: Why It Exists

The Central Question

Josh Naylor stole 30 bases in 2025 at a 93.8% success rate. Baseball Savant ranks him in the 3rd percentile for sprint speed — meaning roughly 97% of major leaguers are faster on their feet. Yet he is one of the most effective base stealers in the game. Standard leaderboards that sort by raw steal totals or raw success rate would never flag him as elite, because they compare him against fast runners operating in a completely different context. The Naylor Model was built to answer a specific question: which runners are systematically outperforming what their speed alone would predict — and why? The answer turns out to involve first-step quickness, lead distance, catcher arm strength, and how well a runner reads specific pitcher deliveries, not top-end speed.

Where the Data Comes From

Everything in this model starts with publicly available Baseball Savant data. Pitch-by-pitch Statcast records (2018–2026, roughly 150,000 to 250,000 pitches per season with a runner on first base) provide the raw event log. From these records the model extracts every stolen base attempt and caught stealing on second base. Sprint speed, jump time, and 5-foot running splits come from Savant's sprint and acceleration leaderboards. Real catcher pop times (2018+) come from Savant's catcher pop-time leaderboard. Pitcher running-game suppression numbers (pickoff rate, step-off rate, hold time) come from Savant's running-game leaderboard. Lead distance is a career-level average per runner, fetched from Savant's base-stealing run value table — note that Savant's year filter on this endpoint is silently ignored; every season query returns the same career snapshot, so lead data is treated as a career constant rather than a per-season number.

The SB Residual: The Core Signal

Why Not Just Use Success Rate?

A runner with a 90% success rate is not automatically better than one with 80%. If the first runner only steals against catchers with slow pop times and pitchers who ignore runners, the 90% might be expected given context. The 80% runner might be attempting far tougher steals. The model's primary signal — called the SB Residual — corrects for this by asking: given this runner's sprint speed, what success rate would we predict for an average runner at that speed, and how much does this specific runner beat or miss that expectation? The residual is simply (real SB%) minus (speed-expected SB%). Positive values mean the runner outperforms their speed peers. Naylor's residual is large and positive. Trea Turner's is smaller than you might expect because his success rate is exactly in line with what a 30 ft/s runner should be able to do.

Bayesian Shrinkage: Why Small Sample Rates Are Distrusted

A runner who goes 1-for-1 has a 100% success rate, but that number is meaningless. The model applies Bayesian shrinkage with a constant of $k=5$: every runner's success rate is computed as $(SB + 5 \times \text{league_average}) / (SB + CS + 5)$. This pulls small-sample rates toward the league mean and only lets extreme rates stand when the sample is large enough. A runner with 30 attempts and 28 successes (93.3%) stays near 93.3% because the sample is real. A runner with 1 attempt and 1 success (100%) gets pulled back toward 78%.

The Three Models — A, B, and the GLM

Why Are There Multiple Models?

The pipeline produces three separate models for different purposes. They are not three versions of the same thing; each answers a different question, and each has a different unit of observation, different features, and different intended use.

Model A: Per-Attempt GBM (the strict one)

Model A is a gradient-boosting machine trained on individual pitch-level steal attempts (roughly 30,000–60,000 rows across 2018–2026). Each row is one specific attempt: this runner, this pitcher, this catcher, this count, this inning. The model tries to predict whether that single attempt succeeded. To prevent data leakage — where the same runner's career-aggregate lead numbers appear in both training and test rows — the model uses 'group cross-validation by runner ID,' meaning a runner's attempts are never split across training and testing. This is the strictest, most honest design. Its AUC is around 0.57–0.59, which looks low. But consider: the baseline success rate in the data is already about 54–58% (managers only send runners on attempts they believe have a good chance), which means a coin flip already gets you 54%. The per-attempt model at 0.59 is barely above that. Individual SB attempts are highly noisy — a bad hop, a perfect slide, a pitcher glancing at the wrong moment — and these are not captured in any data source we have.

Model B: Season-Level GBM (the better performer)

Model B aggregates all of a runner's per-attempt features up to the season level and then asks: did this runner have a ABOVE-average success rate this season? One row per runner-season. A runner who attempted 25 steals and succeeded 22 times (88%) is a 'yes'; one who attempted 20 and succeeded 13 (65%) is a 'no.' By smoothing over a full season of attempts, the noise largely cancels out, and the real signal — is this person actually good at stealing — becomes clearer. Model B's AUC is 0.662–0.700 depending on the era. This is the model whose performance we report as the headline number.

The GLM: Plain-English Weights

The GLM (Simple Logistic Regression) is not designed to be the best predictor. It is designed to be the most interpretable. Unlike the GBMs (which learn thousands of nonlinear interactions between variables), the GLM has one coefficient per feature, and those coefficients translate directly into the 'SB% Boost per Tier' and 'Odds Multiplier' columns you see in the model output. If you want to know whether catcher pop time matters more than jump time, read the GLM weights — the GBM will give a better prediction but won't tell you which feature drove the prediction.

The Baseline P(Success): What Does 62% Mean?

The Intercept as a Starting Point

Every logistic model has an intercept — the predicted probability when all features are exactly at their league-average values. In the Naylor Model this is called the Baseline P(success), and it is approximately 60–62%. That number means: an imaginary runner with exactly average sprint speed, exactly average jump time, exactly average lead distance, facing exactly average catcher pop time and average pitcher suppression, would be expected to succeed on about 62% of steal attempts. This is higher than the naive 50/50 because managers already filter out the bad attempts — they tend to send runners only when the matchup looks favorable.

How the Boost Numbers Work

The 'SB% Boost per Tier' column tells you how much the model's prediction shifts when you improve a single feature by one standard deviation (roughly one tier on the chart). Start at the baseline 62%. If you improve Jump Time from 'average' to 'above average' (one tier better = lower seconds), the model raises its prediction by roughly +10 percentage points to about 72%. If the runner also has above-average lead distance (+4 pp), the model is now at about 76%. These boosts are approximate and additive near the baseline; they compress at extreme probabilities because success rate can't exceed 100%. Negative boost values mean the feature correlates with LOWER success when it improves — this almost never happens except when a feature is in seconds (where lower = better, so a 'tier 1 increase' actually means slower, which hurts).

AUC: Why 0.66-0.70 Is Actually Good

What AUC Measures

AUC (Area Under the ROC Curve) measures how well a binary classifier separates the 'yes' and 'no' outcomes. An AUC of 0.50 means the model is no better than a coin flip. An AUC of 1.00 means perfect prediction. The Naylor Model v6 achieves AUC of about 0.66 on the full 2018–2026 dataset, 0.70 on the pre-2023 era, and 0.68 on the post-2023 era.

Why the Ceiling Is Real, Not a Bug

Stolen-base success has a large irreducible noise component. The throw, the catcher's grip on a given day, whether the pitcher noticed the runner leaning, the umpire's safe/out call on a bang-bang play — none of these appear in any public data feed. Academic baseball research generally finds prediction ceilings around 0.68–0.72 for SB success even with the best features. The Naylor Model is operating at that ceiling. Pushing higher would require private TrackMan or Hawkeye delivery-timing data that teams have internally but do not publish. The pre-2023 era's higher AUC (0.70) is because the signal is cleaner: the 2023 rule changes (larger bases, pickoff limits) structurally shifted the success rate league-wide, and the model needed to adapt to a new regime.

The SSSI: Designing for the Naylor Archetype

What Is the SSSI?

The Slow-Steal Skill Index (SSSI) is not a model prediction — it is a composite scouting score. It weights eight features in a way specifically designed to find runners who outperform their speed: high SB residual, high Accel Gap (first-step quickness relative to top speed), good lead distance, elite catcher arm exploitation, and strong pre-release velocity. Each feature is z-scored (converted to standard deviations from the mean) so they are all on the same scale before weighting.

How Weights Were Chosen Without Overfitting

The SSSI weights were chosen by a grid search — testing thousands of weight combinations and picking the one that best separates elite stealers from average runners in historical data. The obvious danger is that we might find weights that work perfectly for Naylor and Soto just because we tested them. To prevent this, Naylor and Soto were excluded from the weight search entirely. Their data was held out. The 80% of remaining runners were used for optimization; the held-out 20% (including Naylor and Soto's seasons) were used only to verify the final result. Naylor 2025 ranking #1 and Soto 2025 ranking #5 in the final SSSI is therefore a genuine out-of-sample result, not a product of having been used in optimization.

SSSI vs Model B: Different Tools

SSSI ranks every runner on a single composite number. Model B predicts whether a runner will have above-average success this season. They overlap but are not the same. Model B will give Freddie Freeman a high score in 2024 because his season stats show real above-average stealing. SSSI will also reward him but additionally weight the fact that he achieved it with well-below-average sprint speed — a harder feat. The SSSI is the 'how impressive is this' score; Model B is the 'will this runner steal successfully this year' prediction.

Known Limitations and Data Quirks

Lead Data Is Not Per-Season

The Savant base-stealing run value table is supposed to support year-by-year queries, but the year filter is silently ignored by the API — every query returns the same career snapshot regardless of what year you request. This means 'lead distance' and 'lead gain' in this model are career averages, not the specific numbers from a given season. For most runners this barely matters (their approach is consistent), but for a runner who dramatically changed their lead strategy mid-career, the per-season comparison is slightly misleading.

The 3-2 Count Artifact

Statcast's pitch-level data records the balls-strikes count at each pitch. Stolen base attempts, however, are not tagged as individual pitch rows — they are embedded in the 'description text' (des field) of the plate appearance's final pitch. As a result, every parsed SB attempt inherits the count from the LAST pitch of the at-bat, not the actual pitch the runner broke on. The 3-2 count is the most common last-pitch count, so 3-2 shows up as having a 1.9% attempt rate while every other count shows near-zero. This is a data artifact, not a real strategic pattern. The '3-2 Count Attempt Share' variable was retained in v6 as a rough proxy for late-count aggressiveness but should not be interpreted as a precise tactical measure.

Pitcher Delivery Time Unavailable

Per-pitcher delivery time from first movement to release is the single most valuable missing variable. Longer deliveries give runners more time to build speed before the catcher receives the ball. This data exists inside MLB's proprietary TrackMan system but is not published. Savant's 'pitch tempo' CSV was tested but found to have a bug: the median_seconds_onbase column equals median_seconds_empty for every pitcher, suggesting the tempo values do not actually condition on base state. The model uses a league-constant delivery proxy (1.30 seconds) instead. Any future version that gains access to real per-pitcher delivery times would likely push AUC significantly above the current 0.70 ceiling.